GSD RESEARCH MISSION PACKAGE

The Source Code Tells the Story gsd-skill-creator Deep Code Review

Full Documentation Web for Every Line of Code

A Living Architecture Study — v1.49.21 (DACP Milestone)

Vision → Research → Mission

Full Pipeline Execution — Storage is Cheap; Understanding is Priceless

Date: March 27, 2026 | Status: Ready for GSD Orchestrator

Activation Profile: Fleet (17+ roles) | Estimated: 8–12 context windows across 4–6 sessions

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STAGE 1 — VISION DOCUMENT

1 The Source Code Tells the Story — Vision Guide

Date: March 27, 2026

Status: Vision Phase — Pre-Execution

Depends On: gsd-skill-creator dev branch (v1.49.21), The Space Between (tibsfox.com/media/The_Space_Between.pdf)

Context: This mission addresses the complete documentation web for gsd-skill-creator — the adaptive learning layer, GSD execution engine, College of Knowledge, DACP protocol, chipset architecture, holomorphic mathematics engine, and safety/cultural sovereignty systems. All 7,451 files. Every line.

1.1 Vision

There is a principle in the GSD ecosystem: *storage is cheap compared to the value of well-organized information, and all paths should lead to further discovery.* A codebase without documentation trails is like a city with no street signs — you can live there for years without ever knowing what neighborhood you are actually in.

The gsd-skill-creator is not merely a CLI tool. It is the Amiga Principle instantiated in TypeScript. It models skills as complex numbers on a holomorphic plane, routes agent communication through deterministic three-part bundles, maintains a College of Knowledge with 42 academic departments, calibrates itself through continuous feedback loops, and governs itself with cultural sovereignty policies that cannot be overridden by credentials or academic exception. The code *is* the theory. The architecture *is* the philosophy. Every file has a reason. Every design decision has a lineage. Almost none of that is currently visible to someone entering the project cold.

This mission produces the full documentation web: inline JSDoc commentary linking every function back to its intellectual origin, a cross-reference map connecting code modules to College departments and research topics, a code genealogy tracing each subsystem through its milestone history, and a navigator layer so that anyone — contributor, student, AI agent, or future maintainer — can enter at any file and follow trails outward until the entire architecture makes sense. No orphan code. No unexplained decisions. Every door opens onto something richer.

The through-line is the same as in *The Space Between*: meaning lives in the connections between nodes, not the nodes themselves. The nodes already exist. This mission builds the connections.

1.2 Problem Statement

1. **The documentation delta is 25+ milestones wide.** The public docs site is frozen at v1.33. The codebase is at v1.49.21. Twenty-five milestones of architectural decisions — DACP, the holomorphic plane, the chipset Northbridge, AMIGA multi-team governance, the College of Knowledge, Heritage Skills cul-

tural sovereignty, the planning-bridge MCP server — exist only in CHANGELOG entries and source comments. No one can reconstruct the design rationale from those alone.

2. **Every entry point is a dead end without cross-references.** A developer encountering `src/plane/activation.ts` sees "Tangent Activation Engine" but has no trail to the unit circle metaphor in *The Space Between*, the $z = r \cdot e^{i\theta}$ skill model, or the v1.49.3 milestone that introduced geometric scoring. The intellectual depth is invisible.
3. **The College departments are undocumented as integration points.** The 42 departments in `.college/departments/` are the intended destination for documentation cross-references, yet no code file currently links outward to a College wing. The bridge exists architecturally but has never been built at the file level.
4. **Safety boundaries and cultural sovereignty policies have no code-level trail.** The OCAP/CARE/UNDRIP four-level classification system, the physical safety warden, and the 18 red-team scenarios cleared for Heritage Skills deployment are critically important — but they live in CHANGELOG prose, not in code comments visible during development. A future contributor could unknowingly erode a hard boundary without ever seeing its rationale.
5. **The DACP fidelity model is architecturally central but undocumented.** The shift from markdown (non-deterministic) to three-part bundles (deterministic) is the central architectural thesis of v1.49+. Understanding why `FidelityLevel` 0–4 exists, why `.complete` markers use atomic writes, and why opcodes map to ISA encodings requires knowledge that currently lives only in conversation context.

1.3 Core Concept

The Documentation Web: Every source file is a node. Every College department, research topic, mathematical concept, and design decision is also a node. This mission installs the edges — hyperlinks, JSDoc tags, cross-reference files, and a machine-readable index — so the graph becomes navigable from any entry point.

```
Source File (any .ts) | |--- @module tag --> docs/architecture/ page
|--- @see tag --> College department wing URL |--- @milestone tag -->
CHANGELOG entry + milestone spec |--- @mathematical tag --> The Space
Between chapter/section |--- @safety tag --> Safety policy document |
v Navigator Index (.planning/docs-index.json) | +--- by-module/ forward
lookup: file -> all links +--- by-concept/ reverse lookup: concept ->
all files +--- by-department/ College index: dept -> code modules +---
by-milestone/ temporal index: milestone -> files changed
```

1.3.1 Module Architecture

```
MODULE 01: Core Observer-Detective Loop src/observation/ +
src/detection/ + src/learning/ + src/calibration/ The six-step lifecycle:
Observe->Detect->Suggest->Apply->Learn->Compose
```

```
MODULE 02: DACP Protocol src/dacp/ Fidelity 0-4, bus opcodes, bundle
assembly, retrospective analyzer
MODULE 03: Chipset Architecture (Northbridge) src/chipset/
(Copper/Blitter/Gastown/Teams/Exec/Integration) src/amiga/
(CE-1/GL-1/MC-1/ME-1/ICD)
MODULE 04: Application Pipeline and Complex Plane src/application/ stages
+ src/plane/ Skill scoring, token budgets, geometric activation, chord
detection
MODULE 05: College of Knowledge and Rosetta Core .college/ (42
departments, cross-reference resolver, calibration)
MODULE 06: Holomorphic Mathematics Engine src/holomorphic/ +
math-coprocessor/ Mandelbrot/Julia, DMD, Koopman operators, skill dynamics
model
MODULE 07: Safety, Security, and Cultural Sovereignty Heritage Skills
warden, OCAP/CARE/UNDRIP, physical safety domains
MODULE 08: CLI, Infrastructure, and Documentation Web Layer src/cli/ +
www/tibsfox/ + GitHub Actions + docs/
```

1.4 Chipset Configuration

```
# docs-web mission chipset v1.0.0 agnus: role: orchestration model:
claude-opus-4-5 responsibilities: - Cross-module dependency analysis -
College department mapping decisions - Safety boundary verification -
Milestone genealogy reconstruction
denise: role: output-rendering model: claude-sonnet-4-5 responsibilities:
- JSDoc generation for all modules - Cross-reference file production -
Navigator index assembly - docs/ page authoring
paula: role: anticipatory-io model: claude-haiku-4-5 responsibilities: -
Shared tag schema generation - File scaffolding and templating - Index
record insertion
gary: role: glue-logic model: claude-sonnet-4-5 responsibilities: -
CHANGELOG mining and milestone extraction - Safety annotation harvesting
- Verification passes
```

1.5 Scope Boundaries

1.5.1 In Scope (v1.0)

- JSDoc commentary for all public functions in `src/` (TypeScript)
- `@module`, `@see`, `@milestone`, `@mathematical`, `@safety` tag schema and application
- Navigator index (`.planning/docs-index.json`) with four lookup dimensions
- College department cross-reference files (one per department linking to relevant source modules)
- Architecture docs in `docs/architecture/` updated through v1.49.21
- DACP protocol formal specification document
- Chipset architecture reference document (Copper/Blitter/Gastown/AMIGA subsystems)
- Safety and cultural sovereignty policy annotations in Heritage Skills code
- Mathematical foundation annotations linking `src/holomorphic/` and `src/plane/` to *The Space Between*

- CHANGELOG-to-code genealogy map (`docs/GENEALOGY.md`)

1.5.2 Out of Scope

- Rust source in `src-tauri/` (separate mission, separate language)
- GLSL shaders in desktop (visual documentation is a distinct discipline)
- Auto-generated `dist/` files
- `www/tibsfox/com/Research/` HTML output files (these are build artifacts)
- Public-facing tibsfox.com site redesign
- New College department content (this mission documents; it does not expand)

1.6 Success Criteria

1. Every public function in `src/` has JSDoc with at minimum a description, `@param`, `@returns`, and at least one `@see` link.
2. Every source module in `src/` has a `@module` block with a plain-English one-paragraph description and its milestone introduction.
3. All 42 College departments in `.college/departments/` have a corresponding cross-reference file listing all `src/` modules that implement or relate to the department's domain.
4. The Navigator Index (`.planning/docs-index.json`) is complete with all four lookup dimensions populated and machine-validated with AJV.
5. Every `FidelityLevel`, `BusOpcode`, and ICD payload type in `src/dacp/` has inline commentary explaining its purpose and linking to the DACP formal spec.
6. All safety boundaries in Heritage Skills code (`GATE`, `ANNOTATE`, `BLOCK`) are annotated with `@safety` tags that link to the cultural sovereignty policy document.
7. The holomorphic mathematics module (`src/holomorphic/`) has annotations linking each algorithm to its corresponding chapter in *The Space Between*.
8. The complex plane module (`src/plane/`) has annotations linking `TaskVector`, `GeometricScore`, and chord detection to the unit circle metaphor and the $z = r \cdot e^{i\theta}$ skill model.
9. `docs/GENEALOGY.md` traces each major subsystem from its first milestone introduction through its current form, with CHANGELOG excerpts and file-path anchors.
10. The `CLAUDE.md` at project root is updated with documentation web entry points, link conventions, and how to navigate the four index dimensions.
11. All new documentation compiles cleanly (TypeScript typedoc passes, JSON indexes validate with AJV, internal hyperlinks resolve).
12. A contributor with no prior GSD knowledge can, starting from any single `.ts` file, follow documentation links to understand: what the file does, why it exists, when it was introduced, which College department it belongs to, and what mathematical or philosophical principles underpin it.

1.7 Relationship to Other Documents

Document	Relationship
The Space Between (tibsfox.com)	Primary mathematical source; chapters are the `@mathematical` link targets
CHANGELOG.md (dev branch)	Source material for GENEALOGY.md and `@milestone` annotations
docs/CORE-CONCEPTS.md	Extended and updated as part of this mission
docs/architecture/*.m	Completed and brought current to v1.49.21
gsd-skill-creator PR #28 review	Informs which architectural decisions need urgent documentation
GSD v1.50 milestone plan	This mission is preparatory — documentation web enables the systematic retrospective
College of Knowledge (.college/)	Destination for all department cross-reference files produced by this mission

1.8 The Through-Line

The Amiga did not win because it had more RAM than the PC. It won because every chip knew exactly what it was doing and the bus between them was designed for flow, not bottlenecks. The documentation web is that bus for the gsd-skill-creator ecosystem. It does not add new functionality. It removes friction. It eliminates the tax that every new contributor, every new AI agent, every future *you* pays when entering the codebase without a map.

The Space Between is the mathematical spine of this ecosystem. The code is the spine instantiated. The documentation web is the myelin sheath — the insulating layer that makes signal transmission fast and reliable. Without it, every impulse has to find its own path through resistance. With it, the architecture conducts at its designed frequency.

Storage is cheap. Understanding is not. This mission makes understanding cheap too.

STAGE 2 — RESEARCH REFERENCE

2 gsd-skill-creator — Research Reference

2.1 How to Use This Document

This reference documents the actual codebase as found in the dev branch at v1.49.21 (commit frozen March 7, 2026). All findings come from direct source inspection: `git clone --branch dev --depth 1 https://github.com/Tibsfox/gsd-skill-creator.git`. File paths, line counts, class names, and architectural patterns are drawn from the live code.

Key source locations:

- `src/` — 80+ subdirectories, TypeScript library and CLI (7,451 total files, 20,600+ tests)
- `.college/` — 42 departments with RosettaConcept files, calibration models, cross-references
- `.claude/skills/` — 13 auto-activating skill files for the Gastown chipset agents
- `math-coprocessor/` — Python MCP server for mathematical computation (GPU, JIT, streams)
- `docs/` — 158+ markdown documentation files
- `www/tibsfox/com/Research/` — 80+ three-letter research topic directories (mission packs)

2.2 Module 01: Core Observer-Detective Loop

2.2.1 Architecture and Files

The six-step lifecycle (Observe → Detect → Suggest → Manage → Auto-Load → Learn/Compose) is distributed across four source directories: `src/observation/`, `src/detection/`, `src/learning/`, and `src/calibration/`.

File	Class / Export	Role in Lifecycle
observation/transcri	TranscriptParser	Streams JSONL Claude Code transcripts; filters sidechain subagent entries
observation/session-(	SessionObserver	Wraps parser; emits structured session start/end events
observation/pattern-:	PatternSummarizer	Compresses raw transcript entries into compact pattern summaries
observation/promotio	PromotionEvaluator	Decides when a recurring pattern crosses the promotion threshold
observation/promotio	PromotionGatekeepe	Hard safety gate before any pattern becomes a skill candidate
observation/determin:	DeterminismAnalyze	Classifies operations as deterministic vs ephemeral for storage tiering
observation/drift-mo	DriftMonitor	Detects pattern drift and demotes skills that no longer match usage
observation/lineage-:	LineageTracker	Maintains parent-child provenance chain for all promoted skills
detection/pattern-an	PatternAnalyzer	Frequency-map analysis; 3+ occurrences threshold for candidacy
detection/suggestion	SuggestionManager	Interactive REPL for accept/defer/dismiss on skill candidates
detection/gsd-refere	GsdReferenceInject	Injects GSD workflow context into generated skill content
learning/feedback-st	FeedbackStore	JSONL persistence for user corrections (`.planning/patterns/feedback.jsonl`)
learning/refinement-(	RefinementEngine	Applies bounded refinements (max 20% content change, 7-day cooldown)
learning/drift-track	DriftTracker	Measures cumulative skill drift across refinement cycles
learning/version-man	VersionManager	Rollback support for skill refinements
calibration/calibrat:	CalibrationStore	Records activation decisions and user outcomes for threshold tuning
calibration/threshol	ThresholdOptimizer	MCC-based threshold optimization; `calculateMCC()` function

2.2.2 Key Design Decisions for Documentation

JSONL as the observation medium: All observations are appended to `.planning/patterns/sess` The `TranscriptParser` uses Node.js `readline` interface for streaming, never loading the full file into memory. This is a deliberate choice for large sessions — document it with `@design Memory-efficient streaming: readline interface over full-file load for sessions.jsonl.`

Sidechain filter: The transcript parser discards entries where `isSidechain === true`. This filters out subagent Task tool invocations that would otherwise pollute the pattern signal. This single-line check is architecturally critical and needs a `@see` link to the Task tool documentation.

The 20% refinement ceiling: Hardcoded in `learning/refinement-engine.ts` as a `DEFAULT_BOUNDED_CONFIG` constant. This is a safety principle, not a configuration option — the number comes from empirical study of skill drift. Needs `@safety` annotation.

MCC for threshold optimization: The calibration system uses Matthews Correlation Coefficient rather than accuracy or F1 because MCC is robust to class imbalance (most activations are negative). This is a non-obvious choice that every reader will question — document with full `@mathematical` annotation linking to the *The Space Between* statistics chapter.

2.3 Module 02: DACP Protocol

2.3.1 Architecture and Files

DACP (Deterministic Agent Communication Protocol) is the central architectural thesis of the DACP milestone (v1.49+): agent-to-agent communication should pass structured three-part bundles (human intent + structured data + executable code), not ambiguous markdown.

File	Export	Purpose
`dacp/types.ts`	Zod schemas + inferred types	Single source of truth for all DACP types. `FidelityLevel` (0–4), `BusOpcode` (9 values), `BundleManifest`
`dacp/bundle.ts`	`CreateBundleOption` layout	Filesystem layout spec; `.complete` atomicity marker; 50KB data / 10KB script / 100KB total limits
`dacp/assembler/`	`BundleAssembler`	Constructs validated bundles from intent + data + code inputs
`dacp/bus/`	`FilesystemBus`	GSD filesystem bus; routes bundles between agents via directory polling
`dacp/fidelity/`	`FidelityNegotiator`	Upgrades/downgrades bundle fidelity based on agent capability declarations
`dacp/schema-generato`	`generateBundleSche`	Produces JSON Schema from Zod types for runtime validation
`dacp/retrospective/`	`RetrospectiveAnaly`	Post-execution analysis; measures fidelity level adoption over time
`dacp/templates/`	Template bundles	Pre-built bundle templates for standard GSD operations (EXEC, VERIFY, REPORT)
`dacp/msg-fallback.ts`	`MsgFallback`	Graceful degradation to markdown when bundle creation fails

2.3.2 Fidelity Level Model

Level	Name	Description & Documentation Target
0	`PROSE`	Markdown-only intent. Equivalent to current agent communication. Link to milestone introduction in CHANGELOG v1.48.
1	`PROSE_DATA`	Markdown + structured JSON data. First step toward determinism.
2	`PROSE_DATA_SCHEM`	Adds JSON Schema validation. Bundles are now self-describing and verifiable.
3	`PROSE_DATA_CODE`	Adds executable scripts. Receiving agent can run the code directly.
4	`PROSE_DATA_CODE_`	Full bundle with test fixtures. Maximum determinism — receiving agent can verify execution.

The `.complete` marker pattern: Every bundle write ends with `writeFile('.complete', '')`. Consumers call `access('.complete')` before reading. This guarantees atomic reads. Link to `src/security/file-lock.ts` which implements the same pattern at the security layer.

2.3.3 Bus Opcodes

The nine `BusOpcode` values (`EXEC`, `VERIFY`, `TRANSFORM`, `CONFIG`, `RESEARCH`, `REPORT`, `QUESTION`, `PATCH`, `ALERT`) map directly to the GSD Instruction Set Architecture encoding. Each needs a `@see` tag linking to `docs/architecture/` ISA reference and the relevant GSD command definition in `.claude/commands/`.

2.4 Module 03: Chipset Architecture

2.4.1 Chipset Subsystems

Subsystem	Source Path	Role and Documentation Priority
Copper	`src/chipset/co`	Pipeline coprocessor. GSD lifecycle event routing, activation modes, compiler for pipeline instructions. Maps to Amiga Copper: runs in sync with display beam, executing instructions at precise timing. HIGH documentation priority.
Blitter	`src/chipset/bl`	Bulk memory operations. Batch skill moves, content transformations. Maps to Amiga Blitter: DMA-driven bulk copy/fill without CPU.
Gastown	`src/chipset/ga`	State manager for convoy/bead/work-item architecture. `StateManager`, `validateChipset()`. The Northbridge.
Teams	`src/chipset/te`	Agent team topology management within the chipset context.
Exec	`src/chipset/ex`	Low-level execution harness for chipset operations.
Integration	`src/chipset/in`	`StackBridge` — cross-subsystem integration layer.

2.4.2 AMIGA Multi-Team Architecture

The `src/amiga/` module implements full multi-team governance for missions at Fleet scale. Five teams with 14 agents:

Team	Prefix	Responsibility
MC-1	Control Surface	Dashboard (`AlertRenderer`, `TelemetryConsumer`, `CommandParser`)
ME-1	Mission Environment	Manifest, `TelemetryEmitter`, Provisioner, `PhaseEngine`, `GateController`, `SwarmCoordinator`, `ArchiveWriter`
CE-1	Commons Engine	`AttributionLedger`, `ContributionRegistry`, `InvocationRecorder`, `TokenArchitecture`
GL-1	Governance Layer	Charter, Weighting Docs, Dispute Records, Rules Engine, Decision Log, Policy Query
M-5	Meta-Mission Gate	`MetaMissionHarness`, `SkillCandidateDetector` (Launch Gate for self-modification)

The ICD (Interface Control Documents) — `icd-01` through `icd-04` — define the formal typed contracts between teams. These are the most critical documentation targets

in the entire codebase: they are the architectural contract layer.

2.4.3 Gastown Skill YAML Files (Agent Skills)

The `.claude/skills/` directory contains 13 agent skill files implementing the Gastown chipset in SKILL.md format:

- `mayor-coordinator` — Northbridge pattern; convoy dispatch via sling; never executes work directly
- `polecat-worker` — Worker agent receiving dispatched work items
- `witness-observer` — Passive telemetry monitor; escalates to mayor on anomalies
- `sling-dispatch` — The message bus between mayor and polecats
- `beads-state` — Work item state management (bead = atomic unit of work in a convoy)
- `gupp-propulsion` — Forward momentum enforcement; prevents convoy stalls
- `nudge-sync` — Soft synchronization signal between asynchronous agents
- `mail-async` — Async message delivery with delivery receipts
- `refinery-merge` — Output merging and deduplication for parallel worker results
- `hook-persistence` — Hook state persistence across context resets
- `done-retirement` — Clean retirement of completed work items
- `runtime-hal` — Hardware abstraction layer for Claude Code vs Codex vs fallback
- `mayor-coordinator/references/gastown-origin.md` — Origin story for all skills; cross-link target

Every skill file needs its `gastown-origin.md` cross-reference updated to link to the AMIGA architecture docs.

2.5 Module 04: Application Pipeline and Complex Plane

2.5.1 Skill Application Pipeline

The `src/application/` module implements the multi-stage pipeline that loads skills into context:

```
Input Context | v [ScoreStage] -- Relevance scoring via embeddings
+ heuristic | v [ResolveStage] -- Scope resolution (user vs
project precedence) | v [ModelFilterStage] -- Filter by model tier
(Opus/Sonnet/Haiku) | v [LoadStage] -- Content loading and reference
resolution | v [BudgetStage] -- Token budget enforcement (2-5| v
[CacheOrderStage] -- Cache tier ordering (5-minute TTL exploitation) | v
Active Skills (injected into context window)
```

Key class: `SkillPipeline` in `src/application/skill-pipeline.ts` (introduced Phase 52). The `PipelineStage` interface and `PipelineContext` type are the extension points — document with `@extensible` tag explaining the stage composition pattern.

2.5.2 Complex Plane Module (src/plane/)

This is the most mathematically significant module in the codebase and the most under-documented. Skills are positioned on the complex plane as $z = r \cdot e^{i\theta}$ where r encodes relevance magnitude and θ encodes the conceptual angle between concrete and abstract.

File	Mathematical Content Requiring Annotation
`plane/types.ts`	`SkillPosition` (*The Space Between* Ch. 3), `TangentContext`, `TangentMatch`
`plane/arithmetic.t`	`computeTangent()`, `pointToTangentDistance()` — pure geometric functions
`plane/activation.t`	`TaskVector` (concrete/abstract components), `GeometricScore`, tangent-line proximity scoring
`plane/signal-class`	Observer signal classification — links observation module to plane positioning
`plane/chords.ts`	`ChordDetector`, `ChordStore` — chord = stable angular relationship between two skill positions
`plane/promotion.ts`	Plane-aware promotion decisions — bridges observation module to complex plane
`plane/observer-bri`	`ObserverAngularBridge` — converts raw observation signals to angular positions
`plane/position-sto`	Persistence for skill positions across sessions

The `TaskVector` is computed as a 2D position: x-axis (concrete signals: file paths, commands, tool calls) and y-axis (abstract signals: conceptual terms, design patterns, architectural terms). This directly implements the real/imaginary axis decomposition from *The Space Between* Chapter 3.

2.6 Module 05: College of Knowledge and Rosetta Core

2.6.1 Directory Architecture

```
.college/ calibration/ -- Universal calibration engine (delta-store,
engine) college/ -- CollegeLoader, DepartmentExplorer,
CrossReferenceResolver cross-references/ -- xref-registry,
dep-chain-validator, xref-resolver departments/ -- 42 departments, each
with: DEPARTMENT.md -- Human-readable department overview *-department.ts
-- TypeScript department class calibration/ -- Department-specific
calibration model concepts/ -- RosettaConcept files organized by wing
try-sessions/ -- JSON try-session definitions
```

2.6.2 The 42 Departments

All 42 departments need College cross-reference files produced by this mission. Priority tier for code documentation linkage:

Priority	Departments	Code Modules That Link
Tier 1	mathematics, statistics, logic, philosophy	`src/plane/`, `src/holomorphic/`, `calibration/`, `src/conflicts/`
Tier 1	coding, technology, computer science	`src/dacp/`, `src/application/`, `src/detection/`
Tier 1	history, culture	Heritage Skills pack, `src/amiga/gl-1/` governance
Tier 2	physics, electronics, cloud-systems	`src/chipset/`, `math-coprocessor/gpu.ts`
Tier 2	music, art, astronomy	`src/holomorphic/` (Julia/Mandelbrot aesthetics), audio engineering module
Tier 3	All remaining 32 departments	Standard cross-reference files with generic skill-creator entry points

2.6.3 Progressive Disclosure Architecture

`CollegeLoader` implements a three-tier progressive disclosure system: Summary (∼3K tokens, always loaded) → Wing (∼12K tokens, on demand) → Deep (∼50K tokens, on request). This is the same pattern used by `SkillPipeline`'s budget stages and is a core ecosystem pattern. Document with `@pattern Progressive disclosure: three reading speeds (glance/scan/read).`

2.7 Module 06: Holomorphic Mathematics Engine

2.7.1 Architecture Overview

`src/holomorphic/` contains the most mathematically sophisticated code in the project. Ten progressive modules (HD-01 through HD-10), 187 tests across 23 test files, zero external math dependencies.

Subdirectory	Content	Mathematical Target (*The Space Between*)
`complex/arithmet`	`add, sub, mul, div, cexp, cpow`, constants `ZERO, ONE, I`	Ch. 3: Complex Arithmetic
`complex/iterate.`	`computeOrbit(), detectPeriod(), classifyFixedPoi`	Ch. 6: Iteration Theory
`renderer/core.ts`	`mandelbrotEscape juliaEscape(), colorMap()`	Ch. 6: Fractal Geometry
`dmd/`	Five DMD variants: standard, control, multi-resolution, physics-informed, bootstrap	Ch. 8: Information Systems Theory
`modules/`	Koopman operator, Extended DMD with dictionary lifting	Ch. 8: Koopman Lifting
`skills/`	`SkillDynamics` — maps skill-creator onto the complex plane	Cross-module: The Space Between ↔ code

2.7.2 math-coprocessor (Python MCP Server)

The `math-coprocessor/` directory is a Python MCP server providing GPU-accelerated computation. Key files: `server.py` (MCP endpoint), `gpu.py` (CUDA/MPS detection), `jit.py` (Numba JIT compilation), `streams.py` (async streaming for long computations), `vram.py` (VRAM management). This is the only Python code in the project and needs a dedicated architecture doc explaining the TypeScript–Python boundary.

2.8 Module 07: Safety, Security, and Cultural Sovereignty

2.8.1 Safety Boundaries Summary

ABSOLUTE SAFETY BOUNDARIES — These cannot be weakened by documentation, configuration, or credential claims.
Cultural Sovereignty (Heritage Skills):

- Level 4 hard block: no academic exception, no credential override, no self-attestation
- UNDRIP Article 31 compliance required for all Indigenous knowledge ref-

erences
- Nation-specific attribution always required; no generic "Indigenous peoples"
- 24-hour suspension commitment for community removal requests

Physical Safety Domains (10): food, plant, tool, medical, structural, fire, chemical, animal, arctic-survival, marine

- GATE rules are mandatory-pass and override all other logic
- ANNOTATE rules add context but do not block

Skill Self-Modification:

- `push.default=nothing` enforced architecturally
- Mandatory staging gate before any promoted skill reaches main branch
- M-5 Launch Gate in AMIGA: `SkillCandidateDetector` must clear before self-modification

2.8.2 Key Safety Files

File	Safety Annotation Target
`src/security/file-lock`	Atomic write pattern; same as DACP `.complete` marker
`src/security/integrity`	Hash-based integrity verification for skill files
`src/security/operation`	Rate limiting for self-modification operations
`src/validation/path-sa`	`assertSafePath()`, `PathTraversalError` — path traversal prevention
`src/validation/yaml-sa`	`safeParseFrontmatter()` — YAML injection prevention
`heritage-skills-pack/` (warden files)	OCAP/CARE/UNDRIP four-level classification implementation

2.9 Module 08: CLI, Infrastructure, and Documentation Web Layer

2.9.1 CLI Command Architecture

The main entry point `src/cli.ts` imports 25+ command modules from `src/cli/commands/`. Key commands requiring documentation:

Command	Documentation Priority
`calibrate`	Links to calibration module + MCC mathematical basis
`simulate`	Links to simulation module + ActivationSimulator
`chip`	Links to chipset architecture docs
`eval`	Links to evaluation framework + test plan
`budget` / `budget-estimate`	Links to token budget theory + DEFAULT_PROFILES
`mcp-server`	Links to planning-bridge MCP server design
`publish`	Links to packaging pipeline + aminet module
`score-activation`	Links to complex plane scoring + GeometricScore
`reload-embedding`	Links to embeddings module + HuggingFace integration

2.9.2 Research Web (www/tibsfox/com/Research/)

The `www/` directory contains 80+ three-letter research topic directories (e.g., `OCN/` for Ocean Compute, `FIG/` for Fox Infrastructure Group, etc.). Each has `mission-pack/` and `research/` subdirectories. This is the physical research web — every module's documentation should eventually link outward to the relevant research topics. This connection is not currently made anywhere in the codebase.

2.10 Safety and Sensitivity Considerations

Area	Type	Documentation Requirement
Cultural sovereignty policy	ABSOLUTE	Every Heritage Skills code path must `@safety` tag with policy doc link
Physical safety temperatures	ABSOLUTE	Safety temperature constants must be documented as non-overridable
Skill self-modification paths	GATE	M-5 gate must be documented as mandatory before any self-modification
Path traversal prevention	GATE	`assertSafePath()` must be documented at every filesystem entry point
YAML injection prevention	GATE	`safeParseFrontmatter()` must be documented at every frontmatter parse
Academic exception claims	ANNOTATE	Document why academic exception is explicitly refused in Heritage Skills

2.11 Source Bibliography

2.11.1 Primary Sources (Direct Code Inspection)

- **gsd-skill-creator dev branch v1.49.21** — github.com/Tibsfox/gsd-skill-creator (inspected March 27, 2026)
- **CHANGELOG.md** — Complete milestone history from v1.0.0 through v1.49.21
- **CLAUDE.md** — Project entry point; tech stack, key file locations, commit conventions
- **docs/CORE-CONCEPTS.md** — Foundational concepts through v1.49.19
- **docs/HOW-IT-WORKS.md** — Six-step lifecycle and execution engine description
- **docs/architecture/*.md** — Architecture overview, layers, data flows, storage, extending

2.11.2 Mathematical Reference

- **Foxglove, M.T.** *The Space Between.* 923 pp. Available: tibsfox.com/media/The_Space_Betwee — Primary mathematical source for holomorphic module, complex plane, and unit circle skill model annotations

2.11.3 Protocol and Standards

- **UNDRIP Article 31** — United Nations Declaration on the Rights of Indigenous Peoples — Referenced in Heritage Skills cultural sovereignty policy
- **OCAP Principles** — First Nations Information Governance Centre (fnigc.ca) —

Ownership, Control, Access, Possession framework
- **CARE Principles** — Global Indigenous Data Alliance — Collective Benefit, Authority to Control, Responsibility, Ethics
- **Zod v4** (zod.dev) — Schema-first type inference pattern used throughout codebase
- **Vitest** (vitest.dev) — Test framework; 20,600+ tests at v1.49.21

STAGE 3 — MISSION PACKAGE

3 Milestone Specification

3.1 Mission Objective

Produce a complete, machine-navigable documentation web for every source file in gsd-skill-creator (dev branch, v1.49.21) by writing JSDoc commentary with four custom tag types, generating a Navigator Index with four lookup dimensions, creating College department cross-reference files for all 42 departments, authoring architecture and formal specification documents for DACP and the chipset, and updating CLAUDE.md and docs/ to bring all documentation current through the DACP milestone. The output must satisfy the condition: a contributor entering at any single `.ts` file can follow documentation trails to understand what the file does, why it exists, when it was introduced, which College department it belongs to, and what mathematical or philosophical principles underpin it.

3.2 Architecture Overview (Wave Diagram)

```
WAVE 0 [Sequential, Haiku] Tag Schema + Navigator Index Schema + File
Inventory Scaffolding
WAVE 1A [Parallel, Sonnet] WAVE 1B [Parallel, Opus] JSDoc: Modules 01-02
JSDoc: Modules 03-04 (Observation, Detection, (Chipset, AMIGA, Application
Learning, Calibration, DACP) Pipeline, Complex Plane)
WAVE 1C [Parallel, Sonnet] JSDoc: Modules 05-08 (College xrefs,
Holomorphic, Safety, CLI)
CAPCOM GATE (Human review of safety annotations)
WAVE 2 [Sequential, Opus] DACP Formal Spec + Chipset Architecture
Reference GENEALOGY.md + Navigator Index Population The Space Between
annotation pass (holomorphic + plane)
WAVE 3 [Sequential, Sonnet] CLAUDE.md update + docs/ current pass AJV
index validation + typedoc compile pass Final verification matrix check
```

3.3 System Layers

1. **Tag Schema Layer** — Custom JSDoc tags (`@module`, `@see`, `@milestone`, `@mathematical`, `@safety`) with machine-readable structure
2. **Source Annotation Layer** — JSDoc applied to all public functions in `src/` (TypeScript)
3. **Navigator Index Layer** — `.planning/docs-index.json` with four dimensions, AJV-validated
4. **College Bridge Layer** — 42 cross-reference files connecting departments to source modules
5. **Architecture Document Layer** — DACP spec, chipset reference, genealogy map

6. **Entry Point Update Layer** — CLAUDE.md, docs/CORE-CONCEPTS.md, docs/HOW-IT-WORKS.md

3.4 Deliverables

#	Deliverable	Acceptance Criterion	Wave
1	Tag schema definition (`docs/TAG-SCHEMA.mc`	All 5 tag types specified with examples and machine format	0
2	JSDoc: Modules 01–02	Every public function has description + `@param` + `@returns` + `@see`	1A
3	JSDoc: Modules 03–04	Every ICD payload type has fidelity annotation; plane types link to *The Space Between*	1B
4	JSDoc: Modules 05–08	All 42 dept cross-ref files exist; safety tags on all warden paths	1C
5	CAPCOM gate sign-off	Human confirms safety annotations are correct and complete	Gate
6	DACP Formal Spec (`docs/DACP-SPEC.md`)	Fidelity model, opcodes, bundle layout, atomicity pattern fully specified	2
7	Chipset Reference (`docs/CHIPSET-REFEF`	All 6 subsystems + AMIGA teams + ICD contracts documented	2
8	`docs/GENEALOGY.md`	Every subsystem traced from introduction milestone to current form	2
9	Navigator Index (`.planning/docs-inc`	4 dimensions populated; AJV validation passes	2
10	CLAUDE.md update	Documentation web entry points and navigation conventions described	3
11	Final verification pass	typedoc compiles clean; all internal hyperlinks resolve; AJV passes	3

3.5 Component Breakdown

Component	Dependencies	Model	Est. Tokens
Tag Schema + Scaffolding	None	Haiku	15K
File Inventory Script	Tag Schema	Haiku	10K
JSDoc: Observation + Detection	File Inventory	Sonnet	60K
JSDoc: Learning + Calibration	File Inventory	Sonnet	40K
JSDoc: DACP (types + bundle)	File Inventory	Sonnet	45K
JSDoc: Chipset (Cop-per/Gastown)	File Inventory	Opus	70K
JSDoc: AMIGA (ICD + teams)	File Inventory	Opus	80K
JSDoc: Application Pipeline	File Inventory	Sonnet	50K
JSDoc: Complex Plane	*The Space Between*, File Inventory	Opus	65K
JSDoc: College Bridge (42 depts)	File Inventory	Sonnet	55K
JSDoc: Holomorphic + math-coprocessor	*The Space Between*, File Inventory	Opus	70K
JSDoc: Safety + Heritage Skills	OCAP/CARE/UN refs	Opus	60K
JSDoc: CLI + infrastructure	File Inventory	Sonnet	35K
DACP Formal Spec document	DACP JSDoc complete	Opus	40K
Chipset Architecture Reference	Chipset JSDoc complete	Opus	50K
GENEALOGY.md	CHANGELOG, all JSDoc	Sonnet	45K
Navigator Index assembly	All JSDoc complete	Sonnet	30K
AJV validation + typedoc pass	All outputs complete	Sonnet	20K
CLAUDE.md + docs update	All outputs complete	Sonnet	25K

Total estimated tokens: ~865K across 4–6 sessions.

3.6 Model Assignment Rationale

Model	Assigned To	% Budget
Opus	Complex plane math (*The Space Between* linkage), AMIGA ICD contracts, Heritage Skills safety annotations, DACP spec, chipset reference, holomorphic module — all judgment-heavy content where annotation errors have architectural consequences	35%
Sonnet	Observation/detection/learning/calibration JSDoc, application pipeline, College bridge files, CLI docs, GENEALOGY.md, Navigator Index assembly, verification passes	55%
Haiku	Tag schema, scaffolding, file inventory script, template generation	10%

4 Wave Execution Plan

4.1 Wave Summary

Wave	Name	Type	Model	Gate
0	Foundation	Sequential	Haiku	None
1A	Core Loop JSDoc	Parallel	Sonnet	None
1B	Chipset + Plane JSDoc	Parallel	Opus	None
1C	College + Safety JSDoc	Parallel	Sonnet	None
∅	CAPCOM Safety Review	Gate	Human	BLOCK
2	Synthesis Documents	Sequential	Opus	None
3	Verification + Publish	Sequential	Sonnet	None

4.2 Wave 0: Foundation

Task	Output	Model
Define custom tag schema	`docs/TAG-SCHEMA.md` with 5 tag types, examples, machine format	Haiku
Generate file inventory	`.planning/docs-index/file-list.json` — all `.ts` files with module path, line count	Haiku
Scaffold Navigator Index	`.planning/docs-index.json` with empty four-dimension structure + AJV schema	Haiku
Clone CHANGELOG facts	`.planning/docs-index/milestone-map.json` — subsystem → first-introduction milestone	Haiku

4.3 Wave 1A: Core Loop JSDoc (Parallel Track A)

Agent: EXEC-CORE (Sonnet). Files: `src/observation/`, `src/detection/`, `src/learning/`, `src/calibration/`, `src/dacp/`.

Required tag applications: `@module` on every barrel `index.ts`; `@milestone` on every class introduction; `@design` on streaming JSONL choice, sidechain filter, 20% ceiling, MCC selection; `@see` links to College/statistics and College/coding departments.

4.4 Wave 1B: Chipset + Plane JSDoc (Parallel Track B)

Agent: EXEC-ARCH (Opus). Files: `src/chipset/`, `src/amiga/`, `src/application/`, `src/plane/`.

Required tag applications: `@mathematical` on all plane module types linking to *The Space Between* chapters; `@see` links between Copper/Blitter/Gastown and their Amiga hardware analogs; ICD payload types annotated with contract-layer explanation; `@extensible` on `PipelineStage` interface.

4.5 Wave 1C: College + Safety JSDoc (Parallel Track C)

Agent: EXEC-COLLEGE (Sonnet). Files: `.college/`, `src/holomorphic/`, `math-coprocessor/`, safety/security files, `src/cli/`, Heritage Skills warden.

Required tag applications: All 42 department cross-reference files created; `@safety` tags on every GATE/ANNOTATE/BLOCK path in Heritage Skills; TypeScript–Python boundary documented in math-coprocessor; CLI commands linked to their underlying module documentation.

4.6 CAPCOM Gate: Safety Annotation Review

Human review required before Wave 2 proceeds. Verify:

- Every Heritage Skills GATE boundary has a `@safety` tag with correct policy link
- Cultural sovereignty Level 4 hard block is annotated in all relevant paths
- Physical safety temperature constants are annotated as non-configurable
- M-5 Launch Gate annotation is present in AMIGA meta-mission code
- No safety annotation has been softened from BLOCK to ANNOTATE

4.7 Wave 2: Synthesis Documents (Sequential)

Agent: AGNUS-SYNTH (Opus).

Document	Content	Model
`docs/DACP-SPEC.md`	Fidelity 0–4 formal spec, opcode table, bundle layout, atomicity pattern, fidelity negotiation protocol	Opus
`docs/CHIPSET-REFEREN`	Copper/Blitter/Gastown/Teams/Exec/Integratio + AMIGA team topology + ICD-01 through ICD-04 contracts	Opus
`docs/GENEALOGY.md`	All 8 subsystems: first milestone → evolution → current form, with CHANGELOG excerpts and file-path anchors	Opus
Navigator Index population	Merge all `@tag` metadata from Wave 1 into four-dimension index	Sonnet
The Space Between annotation pass	Verify holomorphic and plane modules have correct chapter citations	Opus

4.8 Wave 3: Verification and Publication

Agent: DENISE-VERIFY (Sonnet).

Task	Pass Criterion	Model
`npm run docs:api` (typedoc)	Zero errors; all `@see` links resolve	Sonnet
AJV Navigator Index validation	Schema passes; all four dimensions populated	Sonnet
Internal hyperlink check	All `#anchor` references in docs/ resolve	Sonnet
CLAUDE.md update	Documentation web section added; navigation instructions complete	Sonnet
docs/ currency pass	All outdated "v1.33" references updated to "v1.49.21"	Sonnet

4.9 Cache Optimization Strategy

- Wave 0 outputs (file inventory, tag schema, milestone map) are shared attributes — compute once, cache for all Wave 1 agents
- *The Space Between* PDF is loaded once in Wave 1B/1C and cached for Wave 2 annotation pass
- CHANGELOG.md is loaded once in Wave 0 and not re-fetched
- Heritage Skills policy documents loaded once in Wave 1C and reused for CAPCOM gate and Wave 2 safety pass

4.10 Token Budget

Wave	Tokens (est.)	Sessions	Model
Wave 0	25K	1	Haiku
Wave 1A	145K	1–2	Sonnet
Wave 1B	215K	2	Opus
Wave 1C	185K	1–2	Sonnet
Wave 2	185K	1–2	Opus
Wave 3	110K	1	Sonnet
Total	**865K**	**8–10**	**Mixed**

4.11 Dependency Graph

```
Wave 0: Tag Schema + File Inventory + Milestone Map |
+----------+----------+ | | | 1A 1B 1C (Core) (Arch) (College) | | |
+----+-----+-----+----+ | CAPCOM Gate | Wave 2 (Synthesis Docs) | Wave 3
(Verify + Publish)
```

5 Test Plan

5.1 Test Categories

Category	Count	Priority	Failure Action
Safety-critical	8	Mandatory	BLOCK
Core functionality	22	Required	BLOCK
Integration	9	Required	BLOCK
Edge cases	7	Best-effort	LOG
Total	**46**		

5.2 Safety-Critical Tests

Test ID	Verifies	Expected Behavior
SC-01	Heritage Skills safety tags present	Every GATE/ANNOTATE/BLOCK path in Heritage Skills warden code has a `@safety` tag with policy document URL. Zero untagged safety paths.
SC-02	Cultural sovereignty never weakened	No `@safety` annotation uses ANNOTATE or LOG for paths documented as BLOCK in source policy.
SC-03	Level 4 hard block annotated	`@safety` tags on Level 4 cultural sovereignty blocks include explicit statement that academic exception and credential override are refused.
SC-04	Safety temperature constants documented as non-configurable	All food/medical safety temperature constants annotated with "non-configurable" and safety warden role.
SC-05	M-5 Launch Gate annotation present	`SkillCandidateDetector` and `MetaMissionHarness` have `@safety` tags referencing the mandatory-pass requirement before self-modification.
SC-06	No GPS/location data for endangered species	No documentation or cross-reference file introduces location data for protected species (Heritage Skills ecological references).
SC-07	Path traversal prevention annotated	`assertSafePath()` and `PathTraversalError` have `@safety` tags at every filesystem entry point.
SC-08	YAML injection prevention annotated	`safeParseFrontmatter()` has `@safety` tags at every frontmatter parse site in `src/storage/`.

5.3 Core Functionality Tests (Selected)

Test ID	Verifies	Expected Behavior
CF-01	Every public function has JSDoc	typedoc passes with zero undocumented public symbols
CF-02	Every `index.ts` has `@module` block	Grep for `@module` in all barrel files: 100% coverage
CF-03	Every class has `@milestone` tag	All 40+ exported classes have the milestone that introduced them
CF-04	DACP Fidelity 0–4 fully annotated	`FidelityLevelSchema` and all five levels have inline docs with use-case examples
CF-05	Complex plane types link to *The Space Between*	`TaskVector`, `SkillPosition`, `TangentMatch` have `@mathematical` tags with chapter citations
CF-06	All 42 College dept cross-ref files exist	`find .college/departments -name 'CODE-MODULES.md' | wc -l` equals 42
CF-07	Navigator Index AJV passes	`ajv validate -s .planning/docs-index-schema.json -d .planning/docs-index.json` exits 0
CF-08	GENEALOGY.md covers all 8 subsystems	Document contains sections for Observer Loop, DACP, Chipset, Plane, College, Holomorphic, Safety, CLI
CF-09	DACP Formal Spec complete	`docs/DACP-SPEC.md` contains fidelity table, opcode table, bundle layout diagram, atomicity section
CF-10	Chipset Reference complete	`docs/CHIPSET-REFERENCE.md` contains all 6 subsystems + AMIGA teams + ICD-01 through ICD-04
CF-11	CLAUDE.md updated with docs web section	CLAUDE.md contains “Documentation Web” section with Navigator Index instructions
CF-12	No broken `@see` links	typedoc emits zero “broken link” warnings

5.4 Integration Tests

Test ID	Verifies	Expected Behavior
IN-01	Observation → College bridge	`.college/departments/statistics/CODE-MODULES.md` links to `src/calibration/threshold-optimizer.ts`
IN-02	Plane → mathematics department	`.college/departments/mathematics/CODE-MODULES.md` links to `src/plane/` and `src/holomorphic/`
IN-03	DACP → ISA cross-reference	DACP formal spec cross-references ISA encoding in `docs/architecture/`
IN-04	Heritage Skills → history department	`.college/departments/history/CODE-MODULES.md` links to Heritage Skills warden and cultural sovereignty policy
IN-05	Navigator Index → by-department lookup	`docs-index.json#by-department/mathematics` returns at least 5 source module paths
IN-06	Navigator Index → by-milestone lookup	`docs-index.json#by-milestone/v1.49.0` returns the DACP bundle and types files
IN-07	GENEALOGY → CHANGELOG anchors	Every CHANGELOG excerpt in GENEALOGY.md has a corresponding version anchor in CHANGELOG.md
IN-08	Holomorphic → *The Space Between* chapters	At least 6 distinct chapter citations across `src/holomorphic/` annotations
IN-09	Document self-containment	A reader entering `src/dacp/bundle.ts` can reach DACP Formal Spec, GENEALOGY.md, and College/coding in $\leq$3 link hops

5.5 Verification Matrix

Success Criterion	Test ID(s)	Status
1. Every public function has JSDoc	CF-01, CF-12	Pending
2. Every module has `@module` block + milestone	CF-02, CF-03	Pending
3. All 42 College dept cross-ref files exist	CF-06, IN-01, IN-02, IN-04	Pending
4. Navigator Index complete and AJV-validated	CF-07, IN-05, IN-06	Pending
5. DACP fidelity model fully annotated	CF-04, CF-09, IN-03	Pending
6. Safety boundaries annotated with `@safety` tags	SC-01 through SC-08	Pending
7. Holomorphic module links to *The Space Between*	CF-05, IN-08	Pending
8. Complex plane types link to unit circle model	CF-05, IN-02	Pending
9. GENEALOGY.md traces all 8 subsystems	CF-08, IN-07	Pending
10. CLAUDE.md updated	CF-11	Pending
11. Documentation compiles cleanly	CF-01, CF-07, CF-12	Pending
12. 3-link reachability from any file	IN-09	Pending

6 Mission Crew Manifest

Role	Agent	Responsibility
FLIGHT	Agnus-Orchestrator (Opus)	Mission direction; CAPCOM gate decisions; safety annotation final approval
PLAN	Paula-Planner (Sonnet)	Wave decomposition; parallel track optimization; cache strategy
SCOUT	Scout-Research (Sonnet)	CHANGELOG mining; *The Space Between* chapter mapping; source verification
EXEC-A	Denise-Core (Sonnet)	JSDoc: Observation, Detection, Learning, Calibration, DACP
EXEC-B	Denise-Arch (Opus)	JSDoc: Chipset, AMIGA, Application Pipeline, Complex Plane
EXEC-C	Denise-College (Sonnet)	JSDoc: College bridge files, Holomorphic, Safety, CLI
CRAFT-MATH	Opus-Math (Opus)	*The Space Between* chapter annotations; holomorphic + plane mathematical accuracy
CRAFT-CULT	Opus-Sovereignty (Opus)	Heritage Skills safety annotations; OCAP/CARE/UNDRIP accuracy
VERIFY	Gary-Verify (Sonnet)	typedoc pass; AJV validation; link resolution; safety annotation audit
INTEG	Sonnet-Integ (Sonnet)	Navigator Index population; cross-module consistency; GENEALOGY assembly
CAPCOM	Human-in-Loop	Safety gate sign-off between Wave 1 and Wave 2; scope confirmation
BUDGET	Paula-Budget (Haiku)	Token budget tracking; session boundary planning
SURGEON	Haiku-Health (Haiku)	Documentation quality degradation detection; incomplete annotation flagging
TOPO	Agnus-Topo (Opus)	Mid-mission restructuring if parallel tracks diverge
ARCH	Opus-Arch (Opus)	Cross-cutting structural decisions; ICD contract documentation
SECURE	Opus-Secure (Opus)	Safety boundary enforcement in documentation; prevents annotation weakening
LOG	Sonnet-Log (Sonnet)	Commit messages; milestone summaries; audit trail

7 Execution Summary

Metric	Value
Activation Profile	Fleet (17 roles)
Source Files in Scope	~1,200 `.ts` files in `src/` with public exports
Total Repository Files	7,451
College Departments	42 (all require cross-reference files)
Custom JSDoc Tags	5 (`@module`, `@see`, `@milestone`, `@mathematical`, `@safety`)
Navigator Index Dimensions	4 (by-module, by-concept, by-department, by-milestone)
Architecture Documents Produced	3 (DACP Spec, Chipset Reference, GENEALOGY)
Safety-Critical Tests	8 (all mandatory-pass / BLOCK)
Total Tests	46
Estimated Token Budget	865K
Estimated Sessions	8–10
Estimated Wall Time	2–3 hours (parallel Wave 1 tracks)
Primary Mathematical Reference	*The Space Between* (tibsfox.com)
Human Gates Required	1 (CAPCOM safety review between Wave 1 and Wave 2)

> *The Amiga did not win by having more. It won by knowing exactly what each part was for, and by making every bus between them a first-class design decision. The source code already knows what it is for. This mission makes that knowledge legible to everyone who enters it. Storage is cheap. Understanding is not. All paths should lead to further discovery.*
>
> — From the gsd-skill-creator Documentation Web Vision, March 2026